REGIONAL AGRICULTURAL SOIL TESTING LABORATORY, MANDYA

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA01	**Report Date:**	15-Oct-2026
Location / District:	Maddur Taluk, Mandya District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Red Loamy Soil (Alfisol)	**Target Agro-Zone:**	Karnataka - Mandya Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**245.5**	kg/ha	280.0 - 560.0	Medium (Slightly Deficient)
Available Phosphorus	**P**	**48.2**	kg/ha	23.0 - 56.0	Medium (Optimal)
Available Potassium	**K**	**195.0**	kg/ha	140.0 - 330.0	Medium (Sufficient)
Soil Reaction (pH)	**pH**	**6.80**	pH scale	6.5 - 7.5	Neutral (Ideal)
Electrical Conductivity	**EC**	**0.38**	dS/m	< 1.00	Normal (Non-Saline)
Organic Carbon	**OC**	**0.62**	%	0.50 - 0.75	Medium (Satisfactory)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 245.5 kg/ha [Medium (Slightly Deficient)]

Available Phosphorus (P): 48.2 kg/ha [Medium (Optimal)]

Available Potassium (K): 195.0 kg/ha [Medium (Sufficient)]

Soil Reaction (pH): 6.80 [Neutral (Ideal)]

Electrical Conductivity (EC): 0.38 dS/m [Normal (Non-Saline)]

Organic Carbon (OC): 0.62 % [Medium (Satisfactory)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Well-balanced neutral soil profile with good cation exchange capacity. Available nitrogen is in the low-to-medium category, while phosphorus and potassium reserves are well matched for irrigated cropping in the Cauvery command zone.

General Nutrient Management Guidance: Apply 10-12 t/ha Farm Yard Manure (FYM). Split nitrogenous fertilizer into 3 splits (50% basal, 25% tillering/vegetative, 25% flowering). Basal dose of DAP and MOP recommended according to crop-specific packages of practice for sugarcane, paddy, and ragi.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*